Data Visualization Lab — Practical No. 4
Power BI - VLOG

SYMBIOSIS INSTITUTE OF TECHNOLOGY



UNDER THE GUIDANCE OF -

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AIML B2

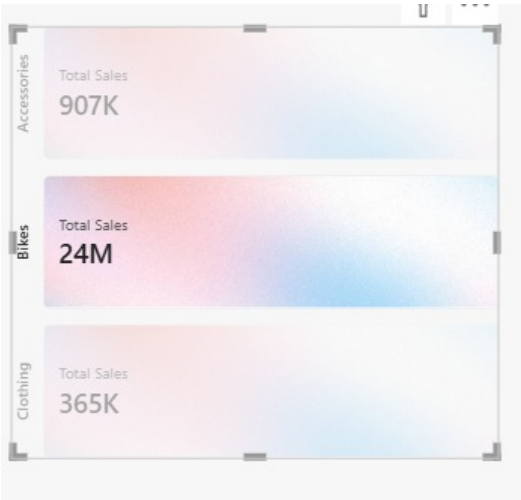
BLOG LINK -

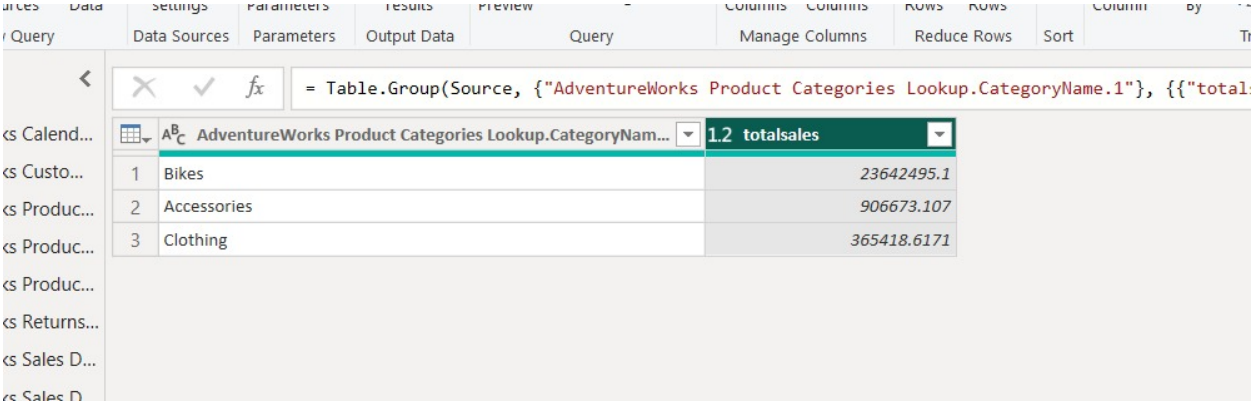
Part A: Power Query Operations

A. Grouping

A1. Total sales amount for each product category

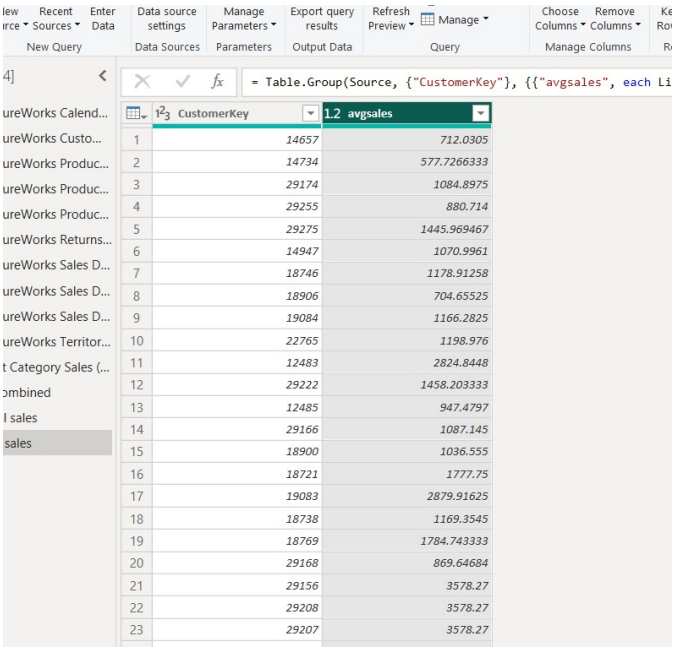
1. Reference Sales Combined → rename 'A1 - Total Sales by Category'.
2. Home → Group By → Group by CategoryName → New column TotalSales, Operation = Sum, Column = SalesAmount.





A2. Average sales amount for each customer

1. Reference Sales Combined → rename 'A2 - Avg Sales by Customer'.
2. Group By → Group by CustomerKey → New column AvgSales, Operation = Average, Column = SalesAmount.



A3. Top 10 customers based on total sales

1. Reference Sales Combined → rename 'A3 - Top 10 Customers'.
2. Group By → Group by CustomerKey → Sum of SalesAmount as TotalSales.
3. Sort TotalSales descending, then Home → Keep Rows → Keep Top Rows → 10.

	CustomerKey	totalsales
1	11433	12407.9545
2	11439	12015.4029
3	11241	11330.4494
4	11417	11085.7504
5	11420	11022.4002
6	11242	10852.034
7	13263	10436.5079
8	12655	10394.9836
9	11425	10391.4304
10	12631	10331.7349

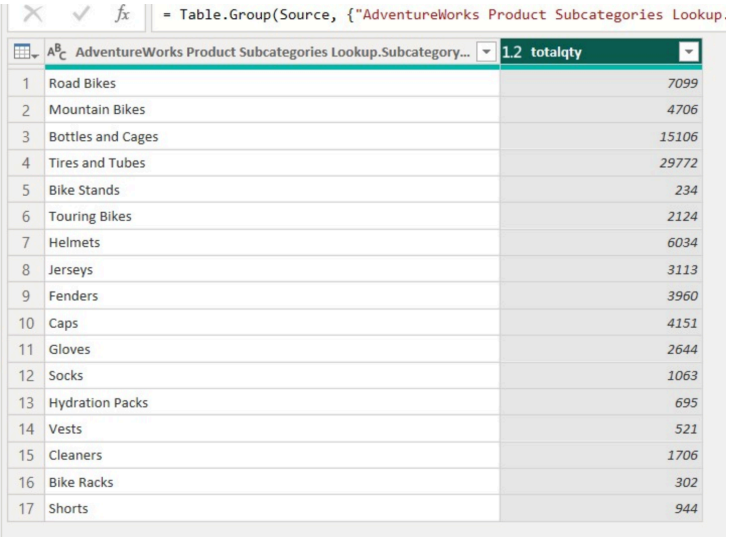
A4. Total quantity sold for each product subcategory

1. Reference Sales Combined → rename 'A4 - Qty by Subcategory'.

2. Group By → Group by SubcategoryName → New column TotalQty, Operation = Sum, Column = OrderQuantity.

Card to add here: Total Qty by Subcategory — pick 4-6 representative subcategories, one card each, measure Total Qty filtered on SubcategoryName.

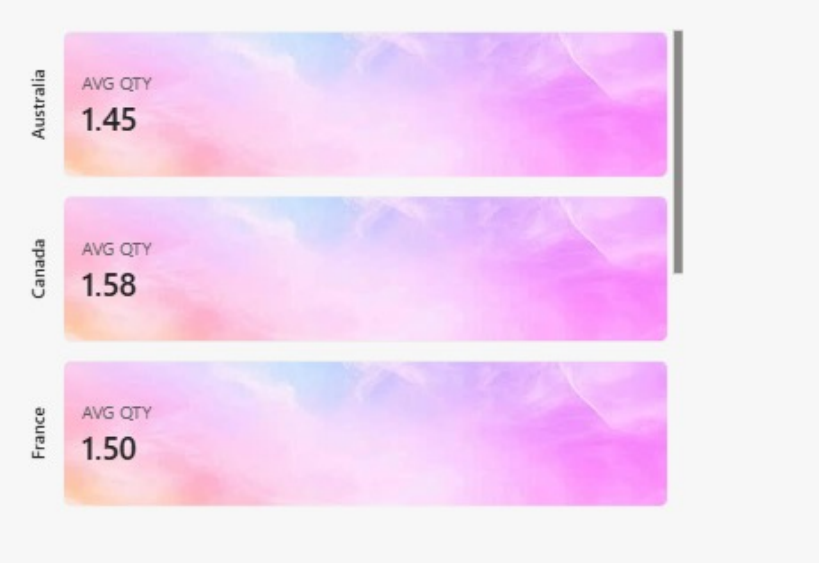




	AdventureWorks Product Subcategories Lookup.Subcategory...	1.2 totalqty
1	Road Bikes	7099
2	Mountain Bikes	4706
3	Bottles and Cages	15106
4	Tires and Tubes	29772
5	Bike Stands	234
6	Touring Bikes	2124
7	Helmets	6034
8	Jerseys	3113
9	Fenders	3960
10	Caps	4151
11	Gloves	2644
12	Socks	1063
13	Hydration Packs	695
14	Vests	521
15	Cleaners	1706
16	Bike Racks	302
17	Shorts	944

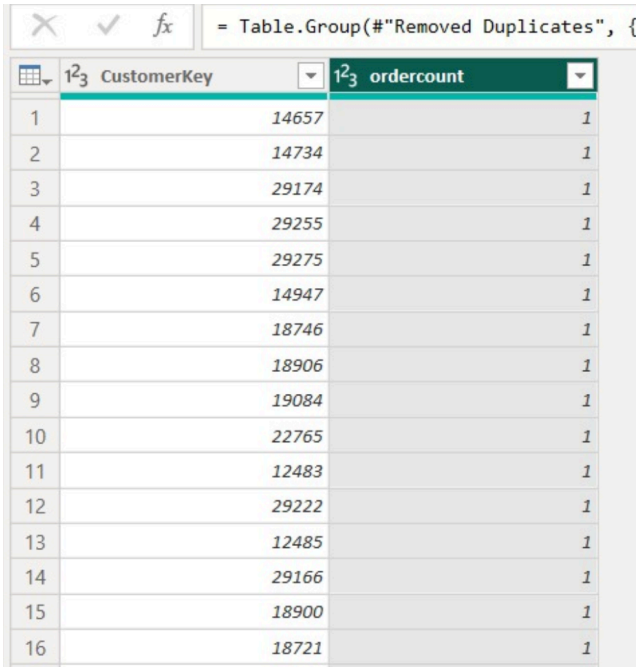
A5. Average order quantity for each sales territory

1. Reference Sales Combined → rename 'A5 - Avg Qty by Territory'.
2. Group By → Group by Country → New column AvgQty, Operation = Average, Column = OrderQuantity.

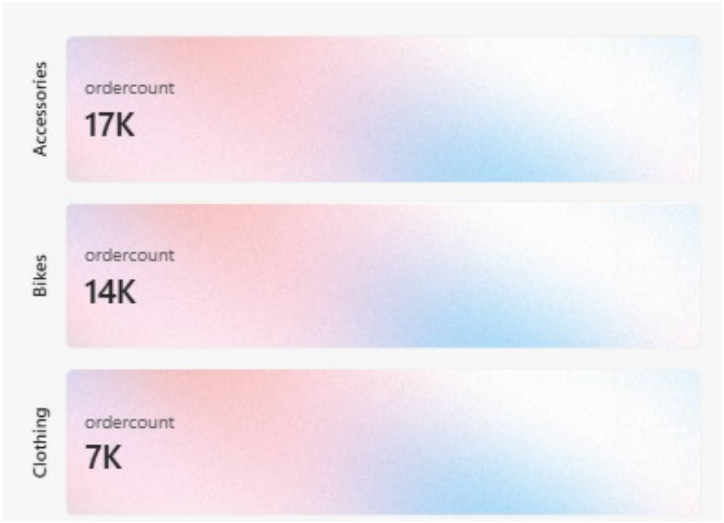


A6. Number of orders placed by each customer

1. Reference Sales Combined → rename 'A6 - Order Count by Customer'.
2. Select CustomerKey and OrderNumber only → Remove Other Columns.
3. Home → Remove Rows → Remove Duplicates (collapses multi-line orders to one row per order).
4. Group By → Group by CustomerKey → New column OrderCount, Operation = Count Rows.

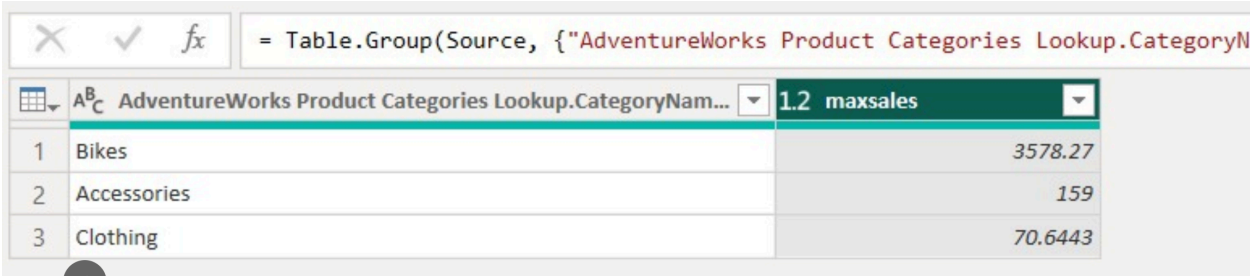


	CustomerKey	ordercount
1	14657	1
2	14734	1
3	29174	1
4	29255	1
5	29275	1
6	14947	1
7	18746	1
8	18906	1
9	19084	1
10	22765	1
11	12483	1
12	29222	1
13	12485	1
14	29166	1
15	18900	1
16	18721	1



A7. Highest sales amount recorded for each product category

1. Reference Sales Combined → rename 'A7 - Max Sales by Category'.
2. Group By → Group by CategoryName → New column MaxSales, Operation = Max, Column = SalesAmount.



	AdventureWorks Product Categories Lookup.CategoryNam...	maxsales
1	Bikes	3578.27
2	Accessories	159
3	Clothing	70.6443

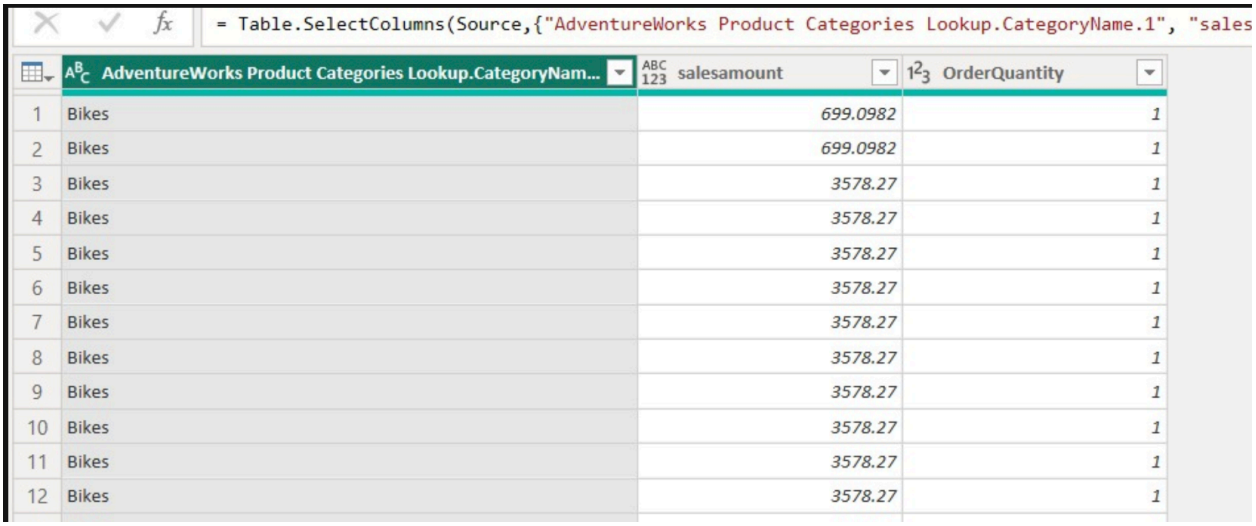
A8. Total tax amount collected for each sales territory

1. No tax column exists in this dataset's source files.
2. Documented as a data limitation, OR added Add Column → Custom Column → EstimatedTax = [SalesAmount] * 0.08 (assumed 8% rate, explicitly flagged as an assumption) → Group By Country, Sum of EstimatedTax

B. Pivoting and Unpivoting

B1. Build base table

1. Reference Sales Combined → rename 'B1 - Pivot Unpivot Demo'.
2. Keep only CategoryName, SalesAmount, OrderQuantity (Remove Other Columns).

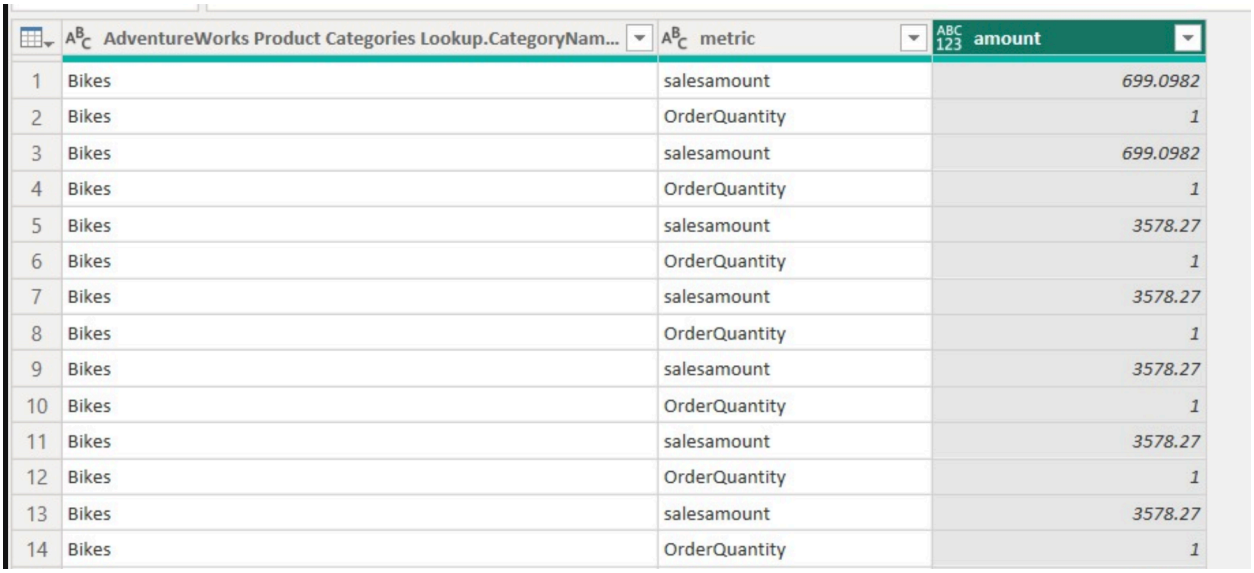


```
= Table.SelectColumns(Source,{"AdventureWorks Product Categories Lookup.CategoryName.1", "sales
```

	AdventureWorks Product Categories Lookup.CategoryNam...	salesamount	OrderQuantity
1	Bikes	699.0982	1
2	Bikes	699.0982	1
3	Bikes	3578.27	1
4	Bikes	3578.27	1
5	Bikes	3578.27	1
6	Bikes	3578.27	1
7	Bikes	3578.27	1
8	Bikes	3578.27	1
9	Bikes	3578.27	1
10	Bikes	3578.27	1
11	Bikes	3578.27	1
12	Bikes	3578.27	1

B2. Unpivot the numerical columns

1. Select SalesAmount and OrderQuantity (not CategoryName).
2. Transform → Unpivot Columns → produces Attribute/Value columns.



	AdventureWorks Product Categories Lookup.CategoryNam...	metric	amount
1	Bikes	salesamount	699.0982
2	Bikes	OrderQuantity	1
3	Bikes	salesamount	699.0982
4	Bikes	OrderQuantity	1
5	Bikes	salesamount	3578.27
6	Bikes	OrderQuantity	1
7	Bikes	salesamount	3578.27
8	Bikes	OrderQuantity	1
9	Bikes	salesamount	3578.27
10	Bikes	OrderQuantity	1
11	Bikes	salesamount	3578.27
12	Bikes	OrderQuantity	1
13	Bikes	salesamount	3578.27
14	Bikes	OrderQuantity	1

B3. Rename the generated columns

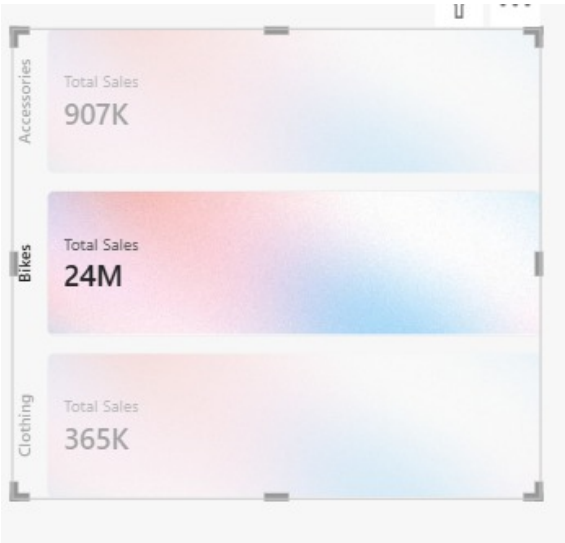
1. Attribute → Metric.
2. Value → Amount.

	AdventureWorks Product Categories Lookup.CategoryNam...	metric	amount
1	Bikes	salesamount	699.0982
2	Bikes	OrderQuantity	1
3	Bikes	salesamount	699.0982
4	Bikes	OrderQuantity	1
5	Bikes	salesamount	3578.27
6	Bikes	OrderQuantity	1
7	Bikes	salesamount	3578.27
8	Bikes	OrderQuantity	1
9	Bikes	salesamount	3578.27
10	Bikes	OrderQuantity	1
11	Bikes	salesamount	3578.27
12	Bikes	OrderQuantity	1
13	Bikes	salesamount	3578.27
14	Bikes	OrderQuantity	1

B4. Pivot the Metric column back

1. Select Metric column → Transform → Pivot Column.
2. Values Column = Amount → Advanced options → Aggregate Value Function = Sum

Attribute	Value
salesamount	699.0982
OrderQuantity	1
salesamount	699.0982
OrderQuantity	1
salesamount	3578.27
OrderQuantity	1
salesamount	3578.27
OrderQuantity	1
salesamount	3578.27
OrderQuantity	1



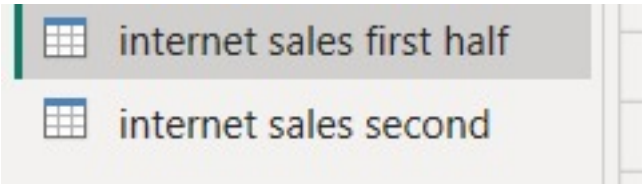
B5. Verify pivoted table matches original

1. Compared SalesAmount totals against A1's Group By result — confirmed match.
2. Spot-checked OrderQuantity against a filtered sum on Sales Combined.

C. Append Queries

C1-C3. First Half / Second Half split and append

1. Referenced Sales Combined twice → 'Internet Sales (First Half)' and 'Internet Sales (Second Half)'.
2. Added Month column (Add Column → Date → Month → Month) to OrderDate on both.
3. Filtered First Half to Month <= 6, Second Half to Month > 6.
4. Home → Append Queries → Append as New → combined into 'Internet Sales (Appended)'.
5. Verified: First Half rows + Second Half rows = Appended rows = 56,046 (matches original Sales Combined total).



D. Merge Queries

D1-D7. Merge Product → Subcategory → Category → Territory

1. Merged Sales Combined with Product Lookup on ProductKey → expanded ProductName, ProductColor, ProductCost.
2. Merged with Product Subcategories Lookup on ProductSubcategoryKey → expanded SubcategoryName.
3. Merged with Product Categories Lookup on ProductCategoryKey → expanded CategoryName.
4. Merged with Territory Lookup on TerritoryKey/SalesTerritoryKey → expanded Region, Country, Continent.
5. Verified no blank/null CategoryName values via the column filter dropdown — every sales record has category info.

Part B: Sales Analysis

Measures used: Total Sales = SUM(SalesAmount), Avg Sales = AVERAGE(SalesAmount), Total Qty = SUM(OrderQuantity), Max Sales = MAX(SalesAmount), Min Sales = MIN(SalesAmount).

1. Total Sales by Product Category

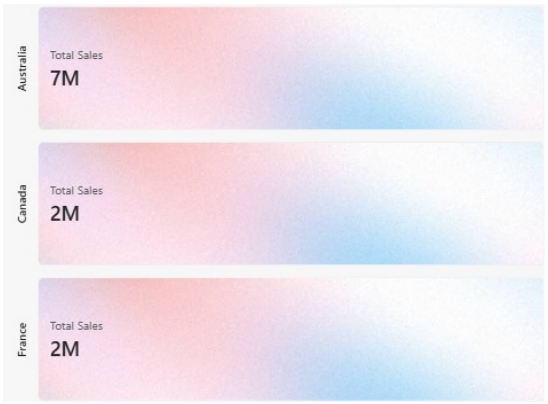
1. Clustered Column Chart or Card set: Axis/Filter = CategoryName, Values = Total Sales.

```
= Table.SelectColumns(Source,{"AdventureWorks Product Categories Lookup.CategoryName.1"
```

	AdventureWorks Product Categories Lookup.CategoryNam...	salesamount	OrderQuantity
1	Bikes	699.0982	1
2	Bikes	699.0982	1
3	Bikes	3578.27	1
4	Bikes	3578.27	1
5	Bikes	3578.27	1
6	Bikes	3578.27	1
7	Bikes	3578.27	1
8	Bikes	3578.27	1
9	Bikes	3578.27	1
10	Bikes	3578.27	1
11	Bikes	3578.27	1
12	Bikes	3578.27	1
13	Bikes	3578.27	1

2. Total Sales by Sales Territory

1. Card set or bar chart: Filter/Axis = Country, Values = Total Sales.



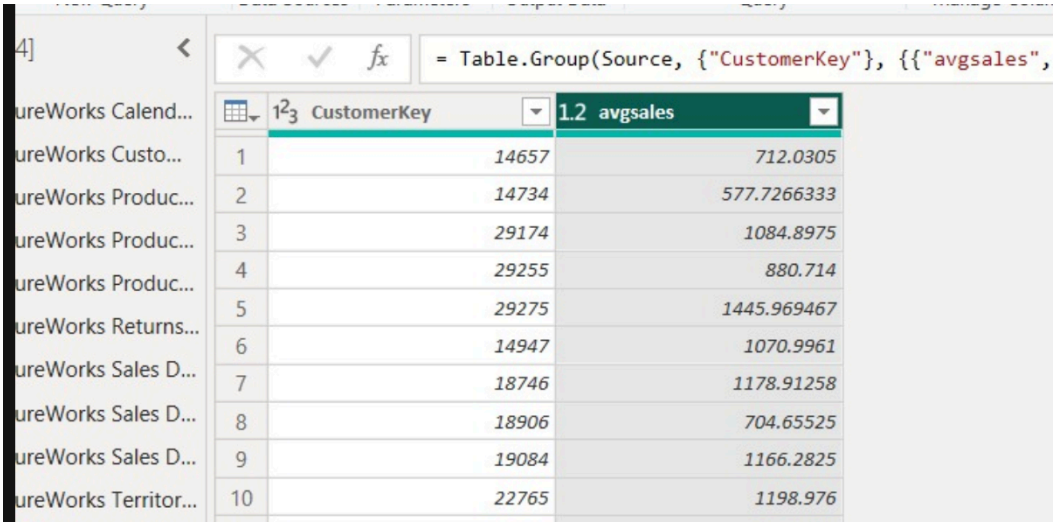
3. Top 10 Products based on Sales Amount

1. Table: ProductName, Total Sales → Filters pane → Top N = 10 on Total Sales.

	AdventureWorks Product Categories Lookup.CategoryNam...	metric	amount
1	Bikes	salesamount	699.0982
2	Bikes	OrderQuantity	1
3	Bikes	salesamount	699.0982
4	Bikes	OrderQuantity	1
5	Bikes	salesamount	3578.27
6	Bikes	OrderQuantity	1
7	Bikes	salesamount	3578.27
8	Bikes	OrderQuantity	1
9	Bikes	salesamount	3578.27
10	Bikes	OrderQuantity	1

4. Top 10 Customers based on Sales Amount

1. Table: CustomerKey (or name), Total Sales → Filters pane → Top N = 10.



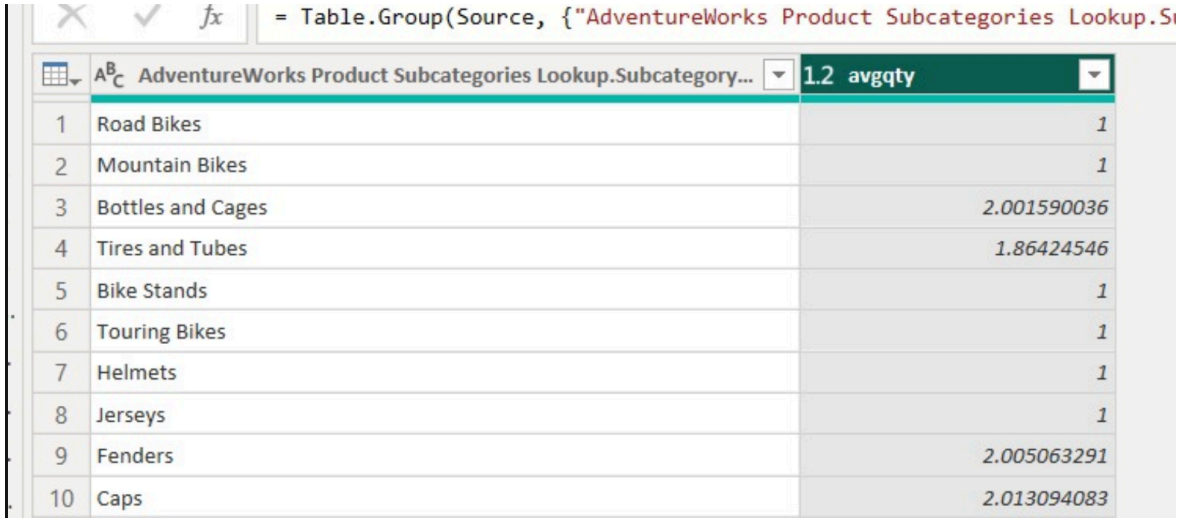
	CustomerKey	avgsales
1	14657	712.0305
2	14734	577.7266333
3	29174	1084.8975
4	29255	880.714
5	29275	1445.969467
6	14947	1070.9961
7	18746	1178.91258
8	18906	704.65525
9	19084	1166.2825
10	22765	1198.976

5. Average Sales Amount by Product Category

1. Card set or bar chart: Filter/Axis = CategoryName, Values = Avg Sales.

6. Total Order Quantity for each Product Category

1. Card set or bar chart: Filter/Axis = CategoryName, Values = Total Qty.

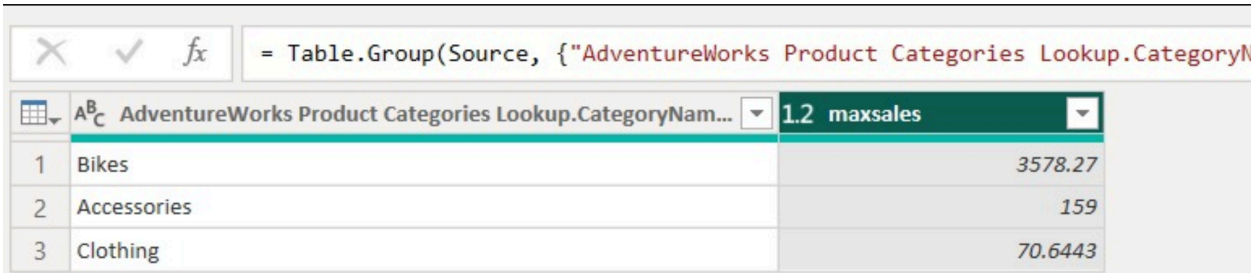


```
= Table.Group(Source, {"AdventureWorks Product Subcategories Lookup.Su
```

	AdventureWorks Product Subcategories Lookup.Subcategory...	1.2 avgqty
1	Road Bikes	1
2	Mountain Bikes	1
3	Bottles and Cages	2.001590036
4	Tires and Tubes	1.86424546
5	Bike Stands	1
6	Touring Bikes	1
7	Helmets	1
8	Jerseys	1
9	Fenders	2.005063291
10	Caps	2.013094083

7. Highest-selling product

1. Card: Max Sales (no filter) for the headline number.
2. Paired Table sorted descending by Total Sales, top row = product name.



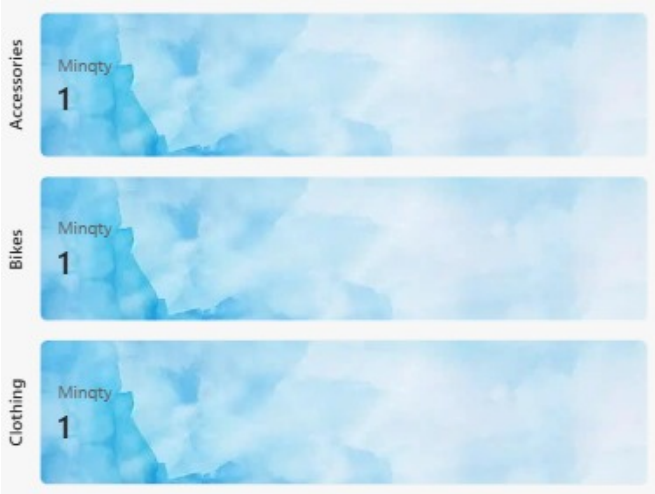
```
= Table.Group(Source, {"AdventureWorks Product Categories Lookup.CategoryN
```

	AdventureWorks Product Categories Lookup.CategoryNam...	1.2 maxsales
1	Bikes	3578.27
2	Accessories	159
3	Clothing	70.6443

8. Lowest-selling product

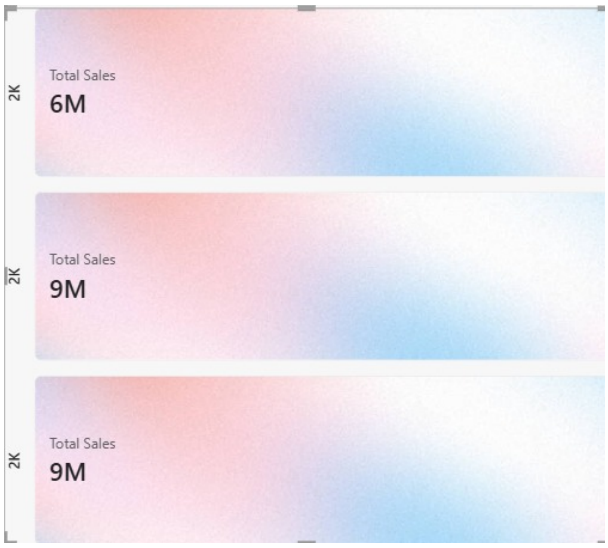
1. Card: Min Sales (no filter).
2. Paired Table sorted ascending by Total Sales, top row = product name.

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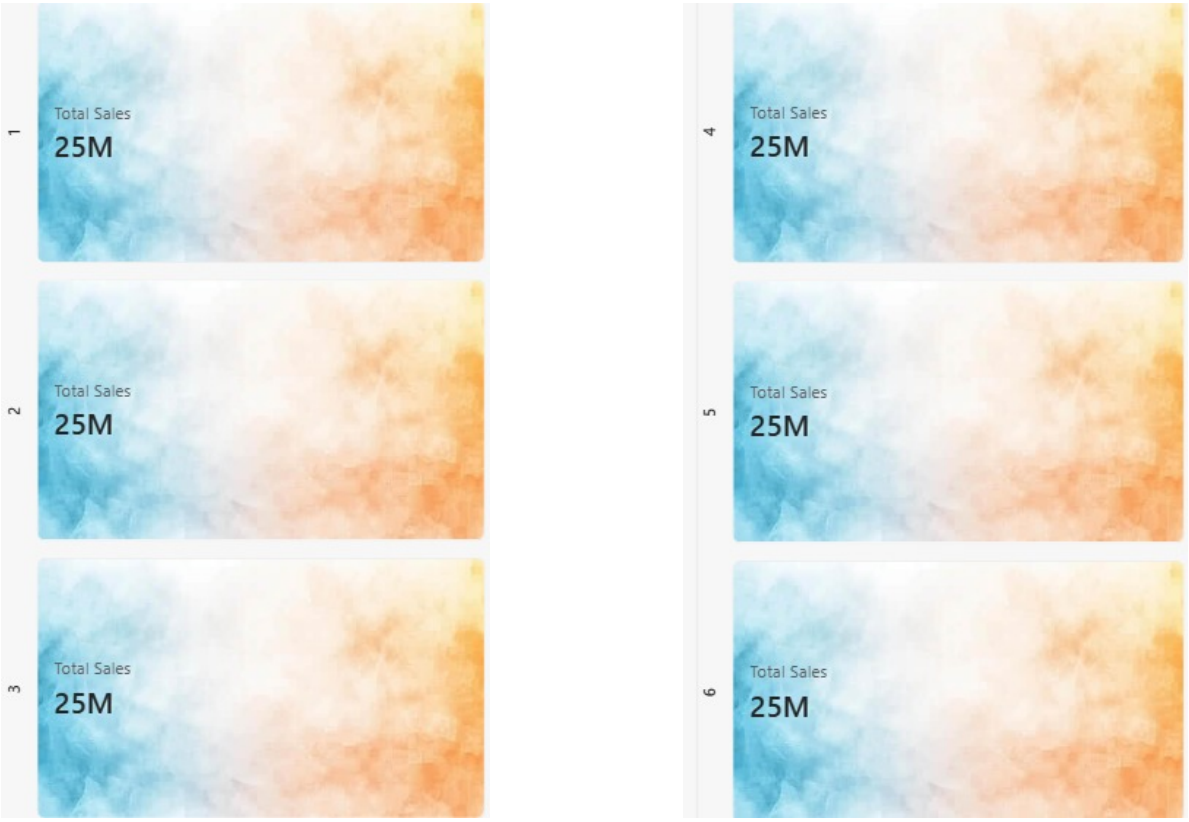
9. Total sales for each calendar year

1. Added Year column (Add Column → Date → Year → Year) on OrderDate.
2. Card set or column chart: Filter/Axis = Year, Values = Total Sales.



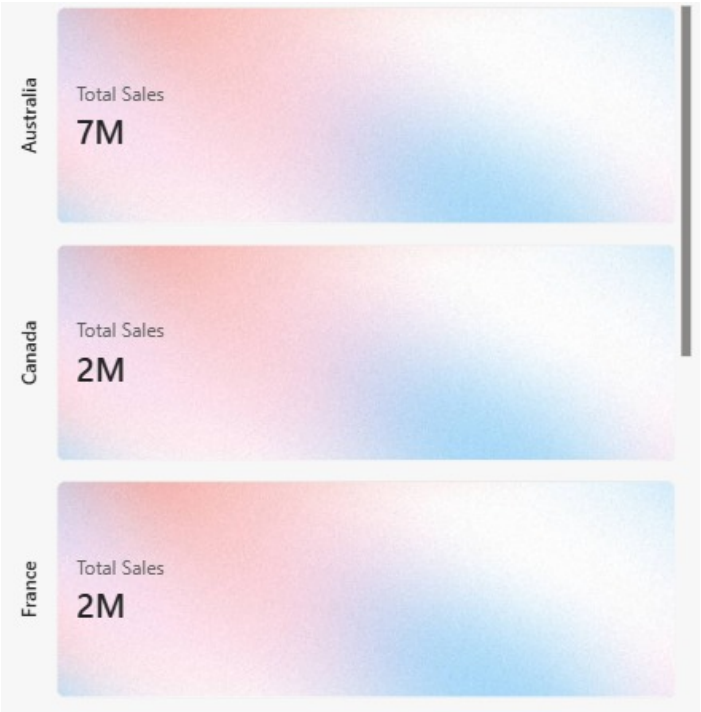
10. Compare sales across different months

1. Added Month Name column on OrderDate.
2. Line chart: Axis = Month Name (sorted by month number), Values = Total Sales.



11. Sales territory with highest revenue

1. Sorted the Territory bar chart / card set from Task 2 descending — top value is the answer.



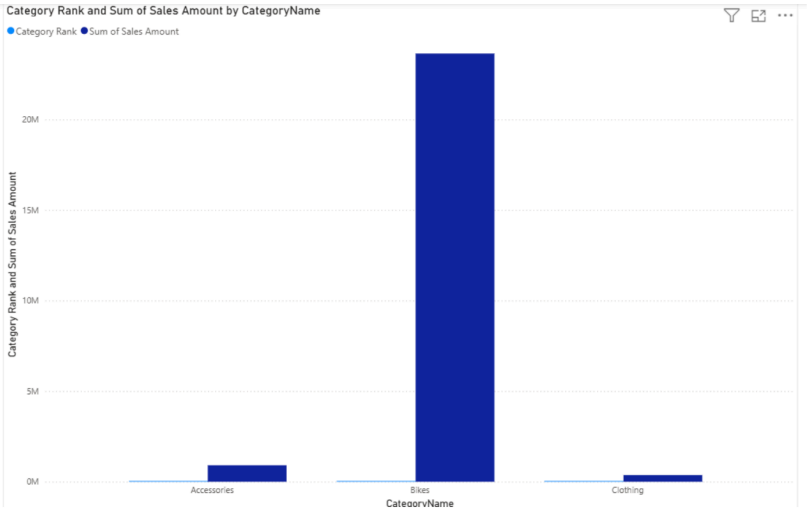
12. Average tax amount collected per territory

1. Only available if the EstimatedTax assumption column (A8) was created.
2. Card set or bar chart: Filter/Axis = Country, Values = Avg Tax = AVERAGE(EstimatedTax).

Advanced Tasks

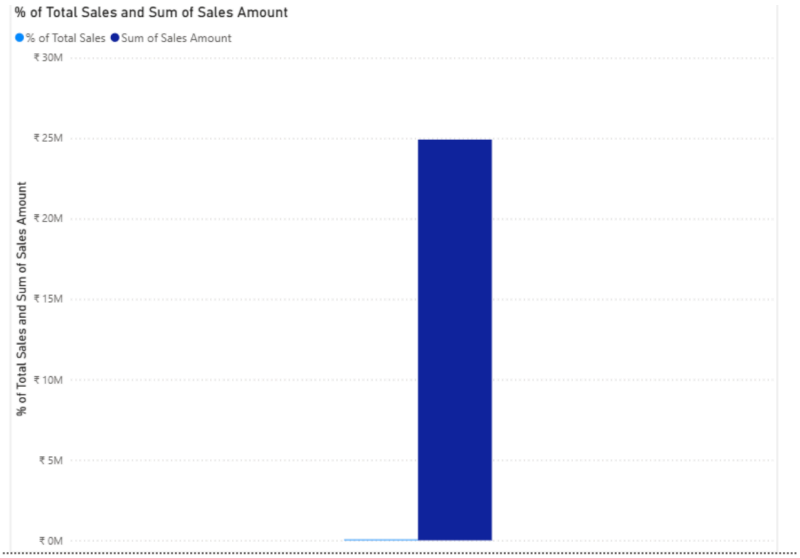
1. Rank product categories by total sales

1. DAX: Category Rank = RANKX(ALL('Sales Combined'[CategoryName]), [Total Sales])
2. Added as a column in the Category table/matrix visual.



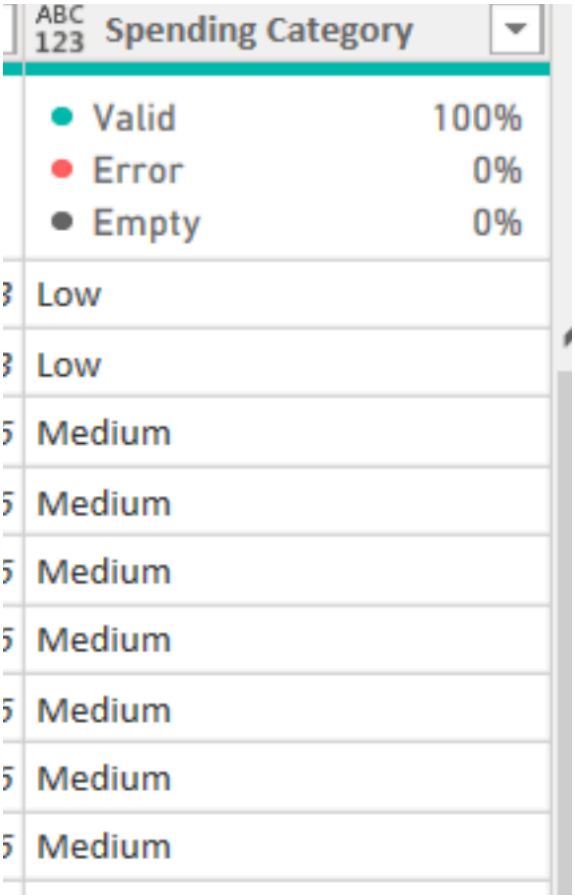
2. Identify products with above-average sales

1. DAX: Overall Avg Sales = AVERAGE('Sales Combined'[SalesAmount])
2. DAX: Above Avg Flag = IF('Sales Combined'[SalesAmount] > [Overall Avg Sales], "Above Average", "Below Average")
3. Filtered a product table to 'Above Average'.



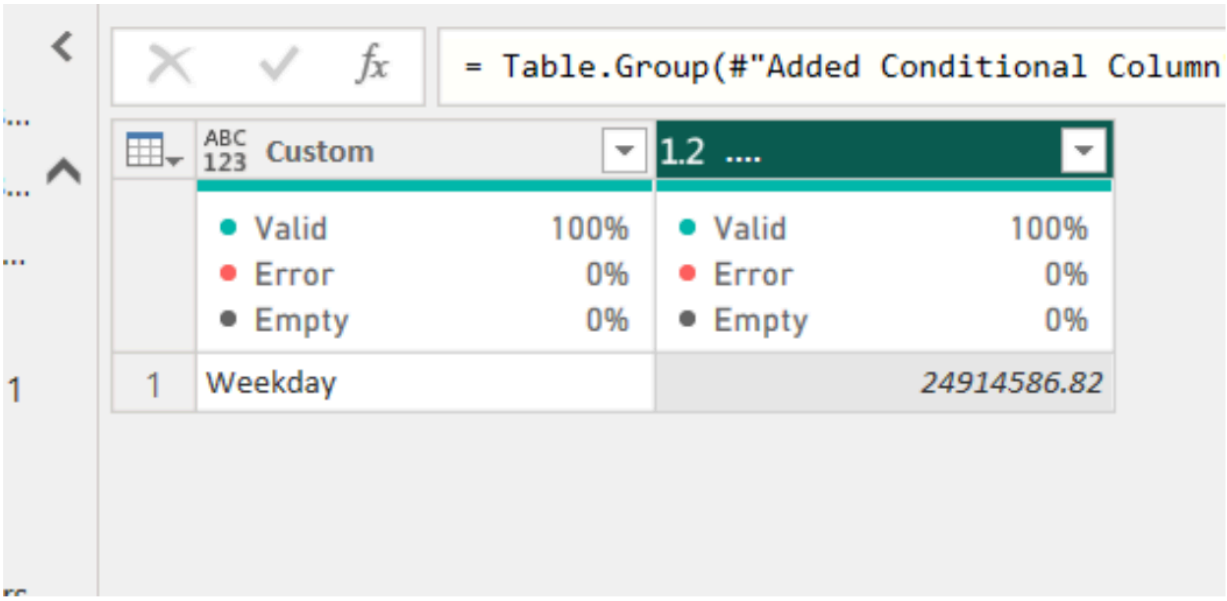
3. Group customers into High / Medium / Low spending (Conditional Column)

1. Grouped CustomerKey with Sum of SalesAmount to get per-customer totals.
2. Add Column → Conditional Column: TotalSales >= 3000 → 'High'; >= 1000 → 'Medium'; else → 'Low' (thresholds tuned to this dataset's distribution — median ~35, 90th percentile ~2070).



4. Percentage contribution of each product category to total sales

1. DAX: % of Total = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL('Sales Combined'[CategoryName])))
2. Formatted as Percentage.



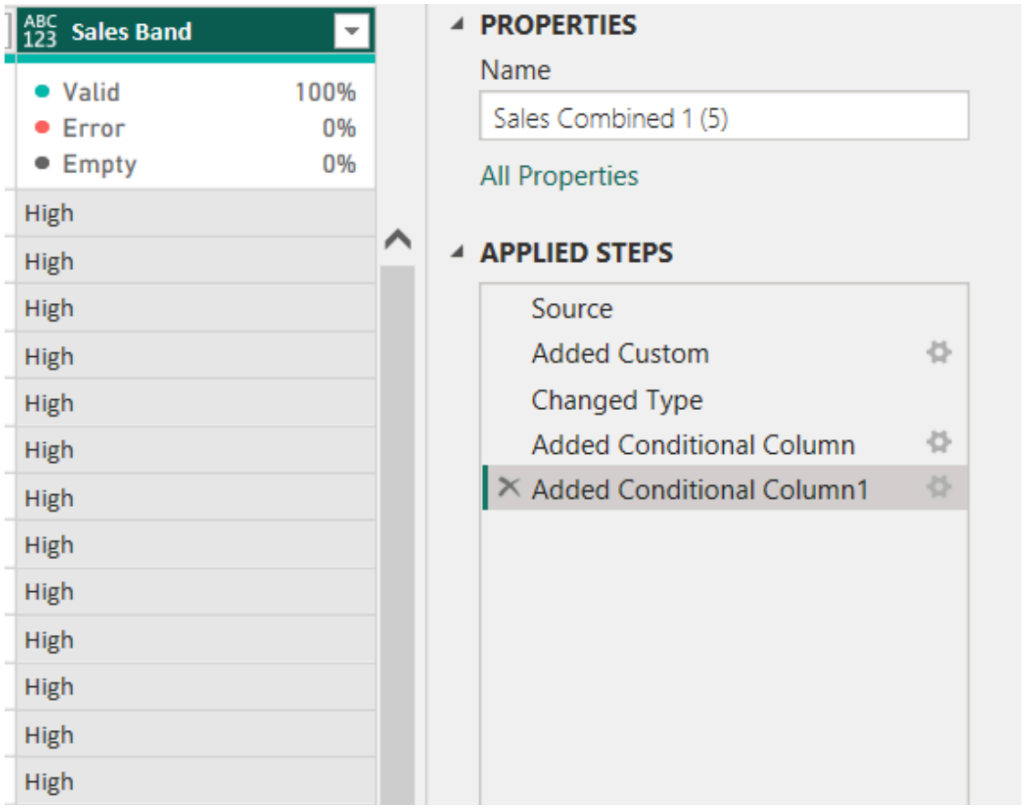
5. Compare weekend vs weekday sales

1. Add Column → Date → Day → Name of Day on OrderDate.
2. Add Column → Conditional Column: Saturday/Sunday → 'Weekend', else → 'Weekday'.

Card to add here: 2 cards — Weekend / Weekday, measure Total Qty or Total Sales filtered on this column.

6. Create sales bands (Low/Medium/High) based on Sales Amount

1. Add Column → Conditional Column on SalesAmount: < 100 → 'Low'; 100-1000 → 'Medium'; > 1000 → 'High' (based on this dataset's actual spread).


Sales Band
Valid 100%
Error 0%
Empty 0%
High
High
High
High
High
High
High
High
High
High
High
High
High
PROPERTIES
Name
Sales Combined 1 (5)
All Properties
APPLIED STEPS
Source
Added Custom
Changed Type
Added Conditional Column
Added Conditional Column1